Penalty Methods for Nonparametric Quantile Regression

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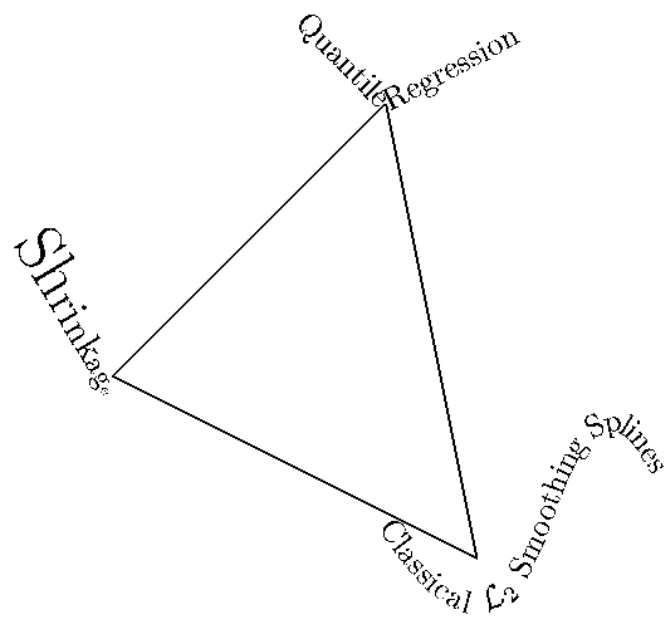
Or ... Pragmatic Goniolatry



"Goniolatry, or the worship of angles, ..."
Thomas Pynchon (*Mason and Dixon*, 1997).

Univariate $\mathcal{L}_2$ Smoothing Splines

The Problem:

$$\min_{g \in \mathcal{G}} \sum_{i=1}^{n} (y_i - g(x_i))^2 + \lambda \int_a^b (g''(x))^2 dx,$$

Gaussian Fidelity to the data:

$$\sum_{i=1}^{n} (y_i - g(x_i))^2$$

Roughness Penalty on $\hat{g}$:

$$\lambda \int_a^b (g''(x))^2 dx,$$

Quantile Smoothing Splines

The Problem:

$$\min_{g \in \mathcal{G}} \sum_{i=1}^{n} \rho_\tau(y_i - g(x_i)) + \lambda J(g),$$

Quantile Fidelity to the Data:

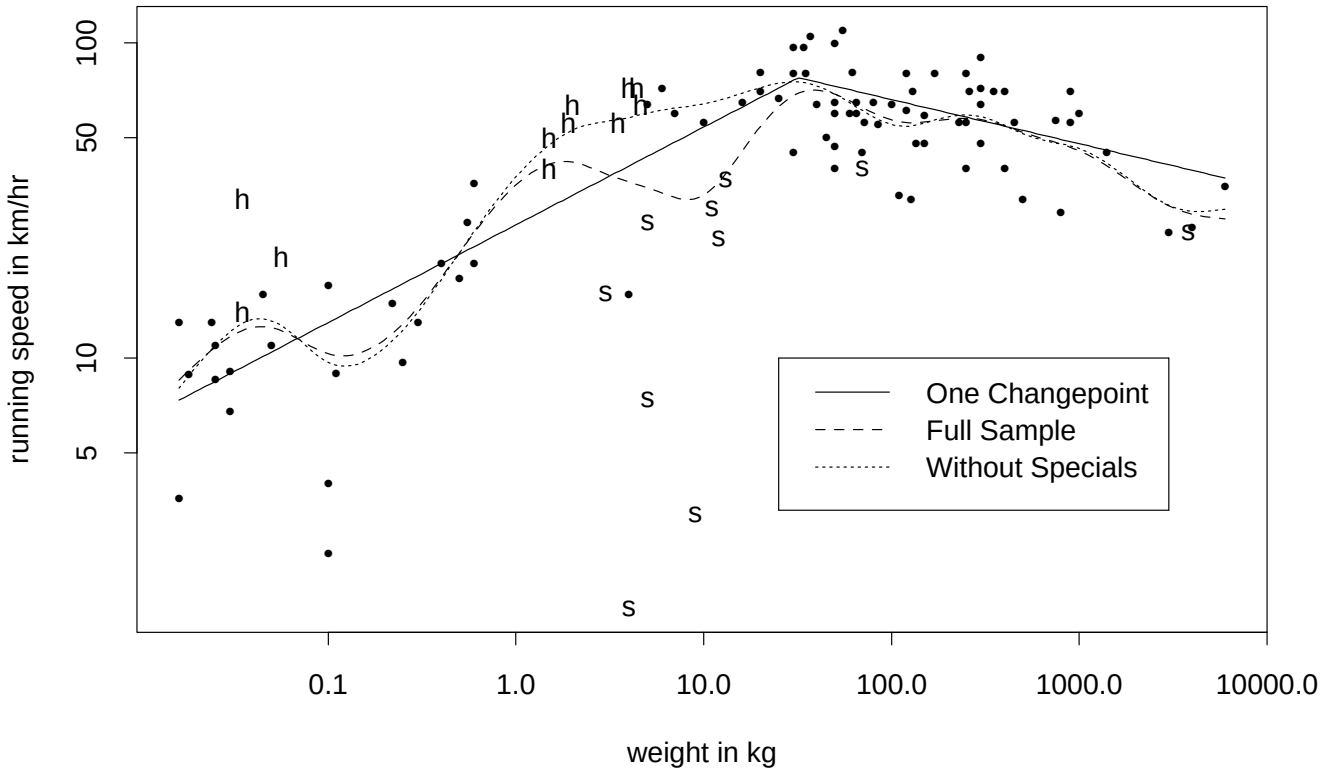
$$\rho_\tau(u) = u(\tau - I(u < 0))$$

Total Variation Roughness Penalty on $\hat{g}$:

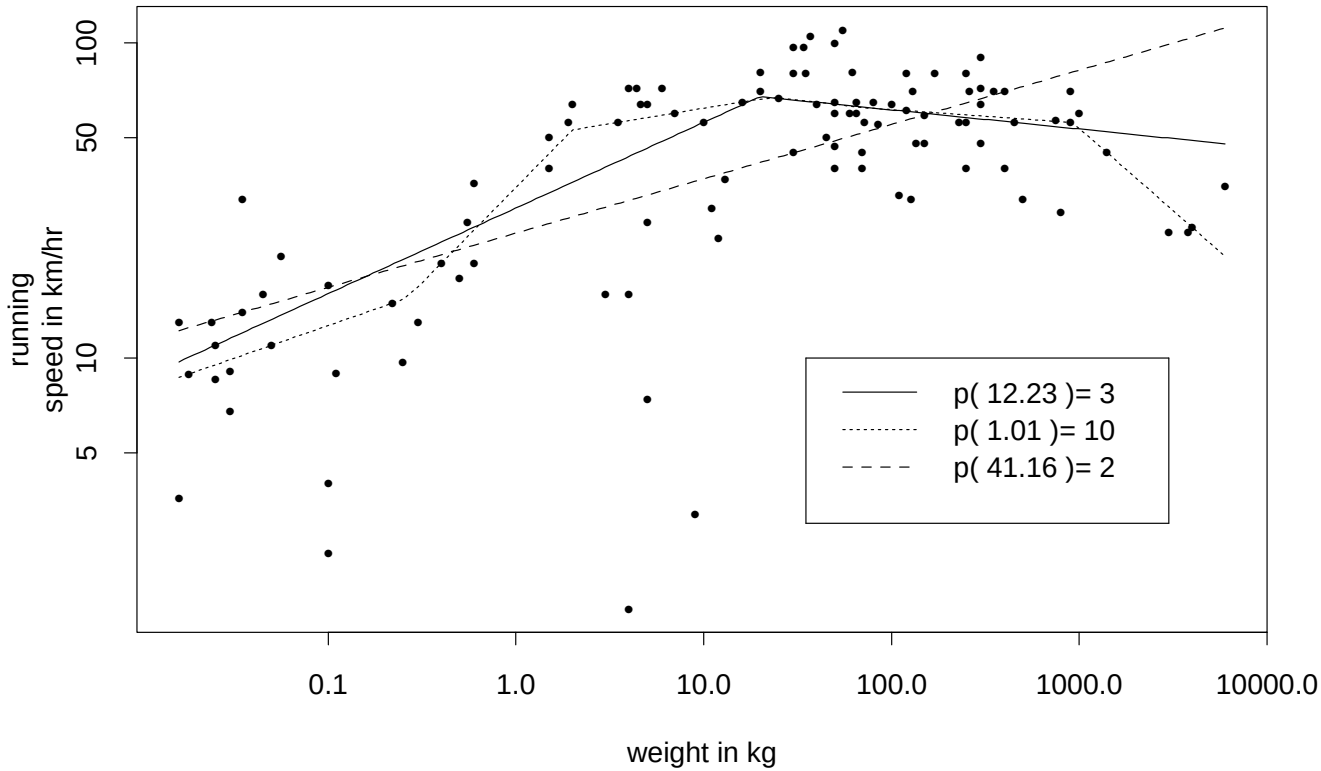
$$J(g) = V(g') = \int |g''(x)| dx,$$

Ref: Koenker, Ng, Portnoy (*Biometrika*, 1994)

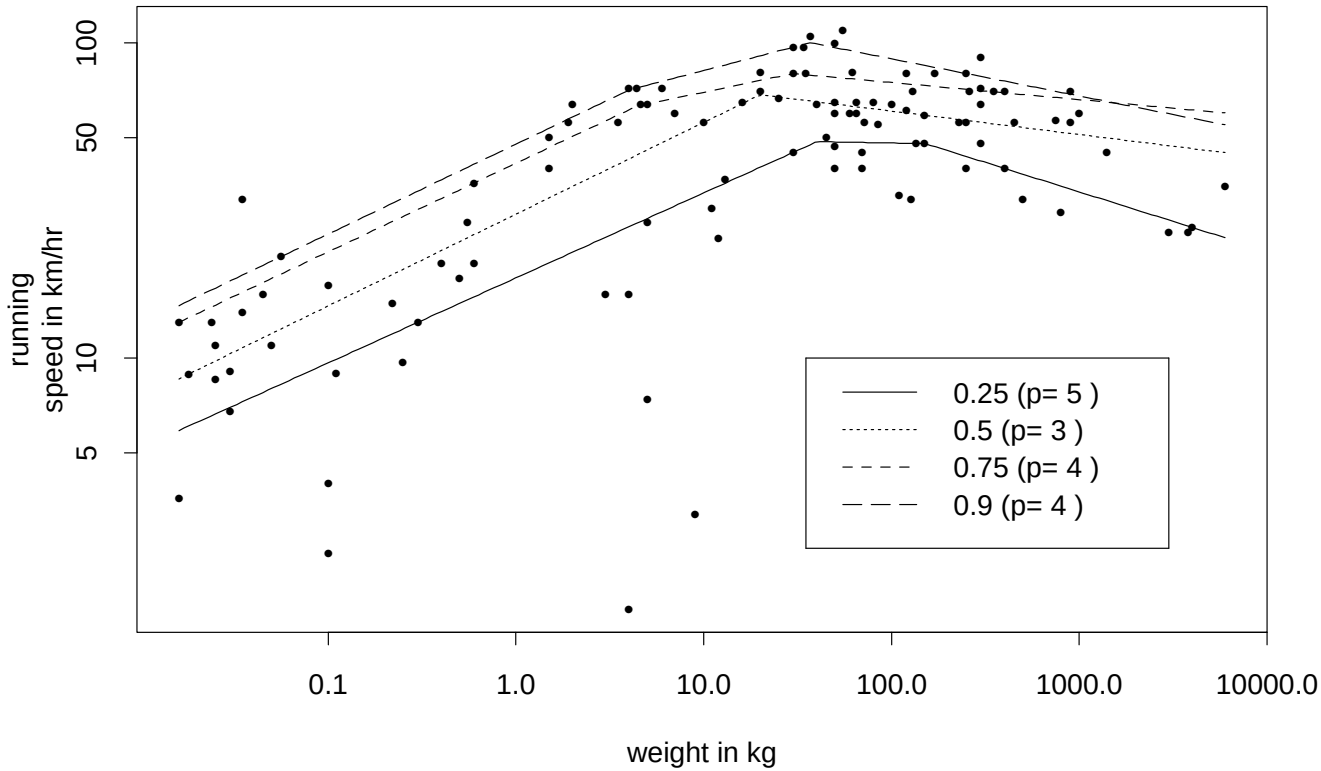
Running Speed of Mammals versus Weight



Three Median Smoothing Spline Fits



Four Quantile Smoothing Spline Fits



Thin Plate Smoothing Splines

Problem:

$$\min_g \sum_{i=1}^{n} (z_i - g(x_i, y_i))^2 + \lambda J(g)$$

Roughness Penalty:

$$J(g, \Omega) = \int\int_{\Omega} (g_{xx}^2 + 2g_{xy}^2 + g_{yy}^2) dx dy$$

Equivariant to translations and rotations.

Easy to compute provided $\Omega = \mathbb{R}^2$. But this creates boundary problems.

References: Wahba(1990), Green and Silverman(1998).

Question: How to extend total variation penalties to $g : \mathbb{R}^2 \to \mathbb{R}$?

Thin Plate Example

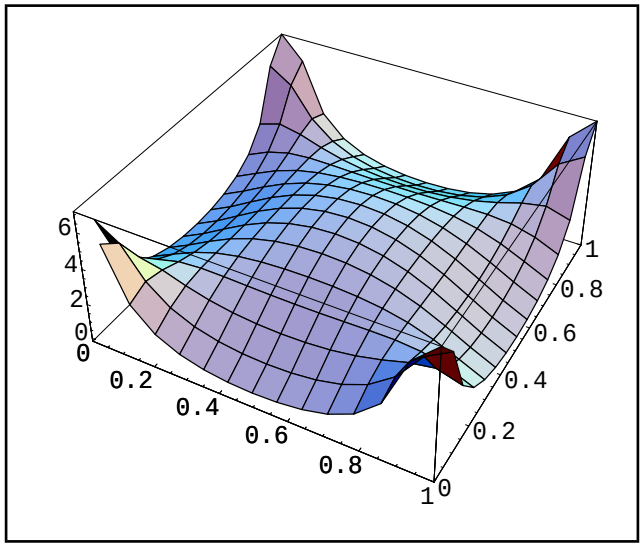


Figure 1: Integrand of the thin plate penalty for the He, Ng, and Portnoy tent function interpolant of the points $\{(0,0,0),(0,1,0),(1,0,0),(1,1,1)\}$. The boundary effects are created by extension of the optimization over all of $\mathbb{R}^2$. For the restricted domain $\Omega = [0,1]^2$ the optimal solution $g(x,y) = xy$ has considerably smaller penalty: 2 versus 2.77 for the unrestricted domain solution.

Three Variations on Total Variation for $f : [a, b] \to \mathbb{R}$

1. Jordan(1881)

$$V(f) = \sup_{\pi} \sum_{k=0}^{n-1} |f(x_{k+1}) - f(x_k)|$$

where π denotes partitions: $a = x_0 < x_1 < \ldots < x_n = b$. ∎

2. Banach (1925)

$$V(f) = \int N(y)dy$$

where $N(y) = \text{card}\{x : f(x) = y\}$ is the Banach indicatrix ∎

3. Vitali (1905)

$$V(f) = \int |f'(x)|dx$$

for absolutely continuous f.

Total Variation for $f : \mathbb{R}^k \to \mathbb{R}^m$

A convoluted history ... de Giorgi (1954)

For smooth $f : \mathbb{R} \to \mathbb{R}$

$$V(f, \Omega) = \int_\Omega |f'(x)| dx$$

- For smooth $f : \mathbb{R}^k \to \mathbb{R}^m$

$$V(f, \Omega, \|\cdot\|) = \int_\Omega \| \nabla f(x) \| dx$$

- Extension to nondifferentiable f via theory of distributions.

$$V(f, \Omega, \|\cdot\|) = \int_\Omega \| \nabla f(x) * \varphi_\epsilon \| dx$$

Roughness Penalties for $g: \mathbb{R}^k \to \mathbb{R}$

For smooth $g: \mathbb{R} \to \mathbb{R}$

$$J(g, \Omega) = V(g', \Omega) = \int_\Omega |g''(x)| dx$$

For smooth $g: \mathbb{R}^k \to \mathbb{R}$

$$J(g, \Omega, \|\cdot\|) = V(\nabla g, \Omega, \|\cdot\|) = \int_\Omega \|\nabla^2 g\| dx$$

Again, extension to nondifferentiable g via theory of distributions.

Choice of norm is subject to dispute.

Invariance Considerations

Invariance helps to narrow the choice of norm.

For orthogonal U and symmetric matrix H, we would like:

$$\|U^\top H U\| = \|H\|$$

- Examples:

$$\| \nabla^2 g\| = \sqrt{g_{xx}^2 + 2g_{xy}^2 + g_{yy}^2}$$
$$\| \nabla^2 g\| = |trace\, \nabla^2 g|$$
$$\| \nabla^2 g\| = max|eigenvalue(H)|$$

-

$$\| \nabla^2 g\| = |g_{xx}| + 2|g_{xy}| + |g_{yy}|$$
$$\| \nabla^2 g\| = |g_{xx}| + |g_{yy}|$$

- Solution of associated variational problems is difficult!

Triograms

Following Hansen, Kooperberg and Sardy (JASA, 1998):

Let $\mathcal{U}$ be a compact region of the plane, and let Δ denote a collection of sets $\delta_i : i = 1, \ldots, n$ with disjoint interiors such that $\mathcal{U} = \cup_{\delta \in \Delta} \delta$.

If $\delta \in \Delta$ are planar triangles, Δ is a triangulation of $\mathcal{U}$,

Definition: *A continuous, piecewise linear function on a triangulation, Δ, is called a triogram.*

For triograms roughness is less ambiguous.

A Roughness Penalty for Triograms

For triograms the "ambiguity of the norm" problem for total variation roughness penalties is resolved.

Theorem. Suppose that $g : \mathbb{R}^2 \to \mathbb{R}$, is a piecewise-linear function on the triangulation, Δ. For any coordinate-independent penalty, J, there is a constant c dependent only on the choice of the norm such that

$$J(g) = cJ_{\triangle}(g) = c \sum_{e} \|\nabla g_e^+ - \nabla g_e^-\| \, \|e\| \tag{1}$$

where e runs over all the interior edges of the triangulation $\|e\|$ is the length of the edge e, and $\|\nabla g_e^+ - \nabla g_e^-\|$ is the length of the difference between gradients of g on the triangles adjacent to e.

Computation of Median Triograms

The Problem:

$$\min_{g \in \mathcal{G}_\triangle} \sum |z_i - g(x_i, y_i)| + \lambda J_\triangle(g)$$

can be reformulated as an augmented ℓ_1 (median) regression problem,

$$\min_{\beta \in \mathbb{R}^p} \sum_{i=1}^{n} |z_i - a_i^\top \beta| + \lambda \sum_{k=1}^{M} |h_k^\top \beta|.$$

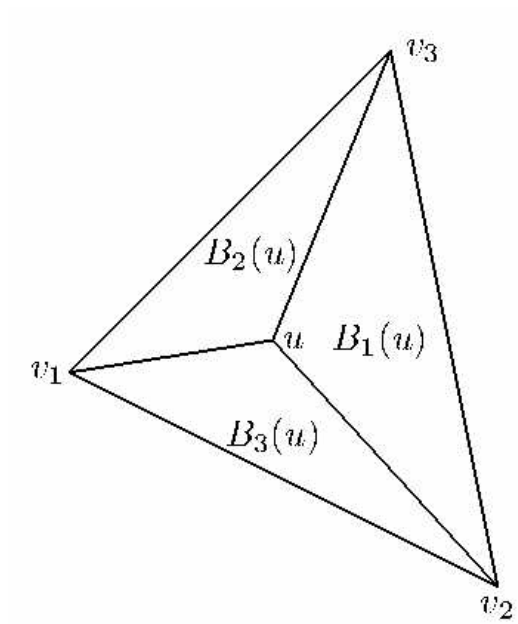
where β denotes a vector of parameters representing the values taken by the function g at the vertices of the triangulation $\triangle$. The a_i are barycentric coordinates of the (x_i, y_i) points in terms of these vertices, and the h_k represent the penalty contribution in terms of these vertices.

Extensions to quantile and mean triograms are straightforward.

Barycentric Coordinates

Triograms, $\mathcal{G}$, on Δ constitute a linear space with elements

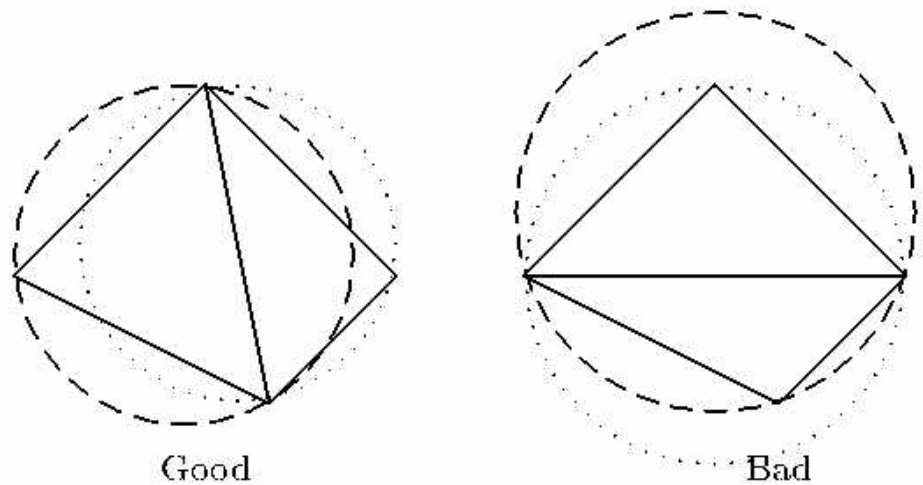
$$g(u) = \sum_{i=1}^{3} \alpha_i B_i(u) \quad u \in \delta \subset \Delta \qquad B_1(u) = \frac{\text{Area } (u, v_2, v_3)}{\text{Area } (v_1, v_2, v_3)} \text{ etc.}$$



Delaunay Triangulation

Properties of Delaunay triangles:

- Circumscribing circles of Delaunay triangles exclude other vertices,
- Maximize the minimum angle of the triangulation.



Robert Delaunay

B.N. Delone (1890-1973)

Four Median Triograms Fits

Consider estimating the noisy cone:

$$z_i = \max\{0, 1/3 - 1/2\sqrt{x_i^2 + y_i^2}\} + u_i,$$

with the (x_i, y_i)'s generated as independent uniforms on $[-1, 1]^2$, and with the u_i are iid Gaussian with standard deviation $\sigma = .02$. With sample size $n = 400$, the triogram problems are roughly 1600 by 400, but very sparse.

Four Median Triograms Fits

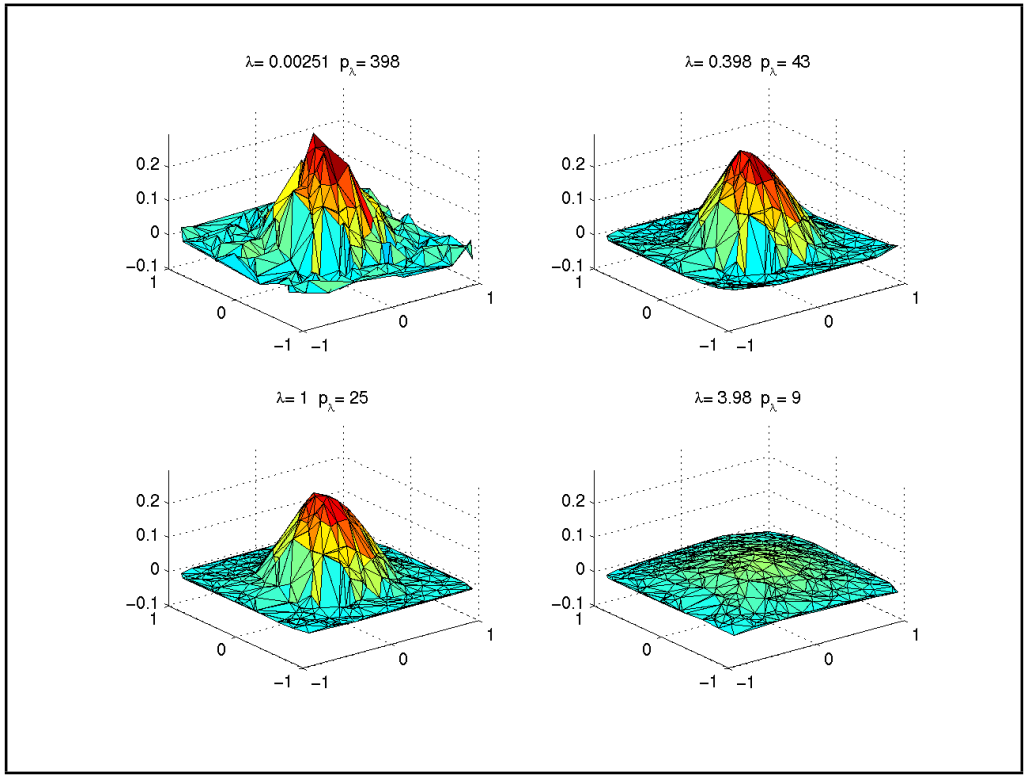
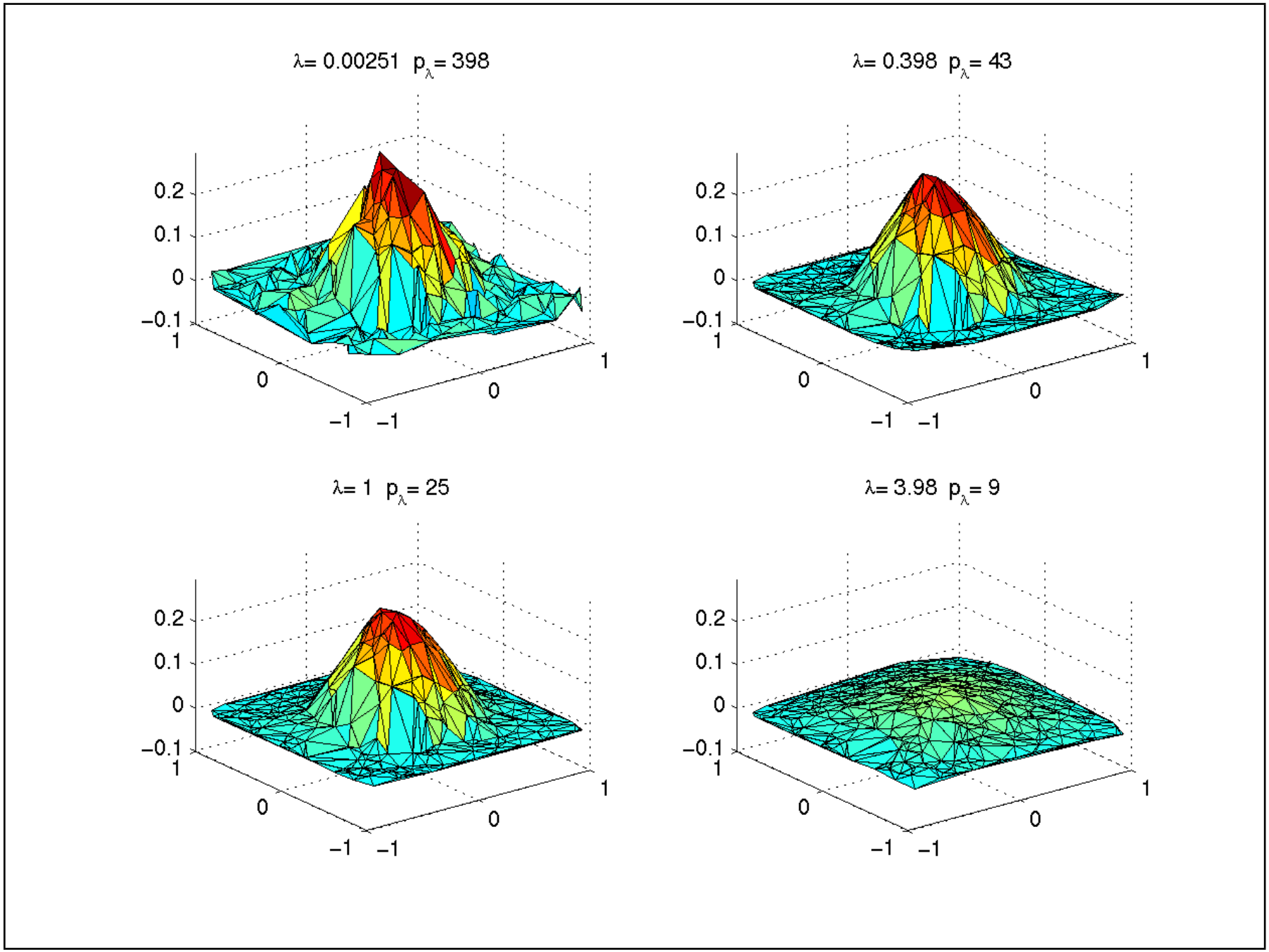


Figure 2: Four median triogram fits for the inverted cone example. The values of the smoothing parameter λ and the number of interpolated points in the fidelity component of the objective function, p_λ are indicated above each of the four plots.


λ= 0.00251 p_λ = 398
λ= 0.398 p_λ = 43
λ= 1 p_λ = 25
λ= 3.98 p_λ = 9
0.2
0.1
0
−0.1
1
0
−1
−1
0
1

Four Mean Triograms Fits

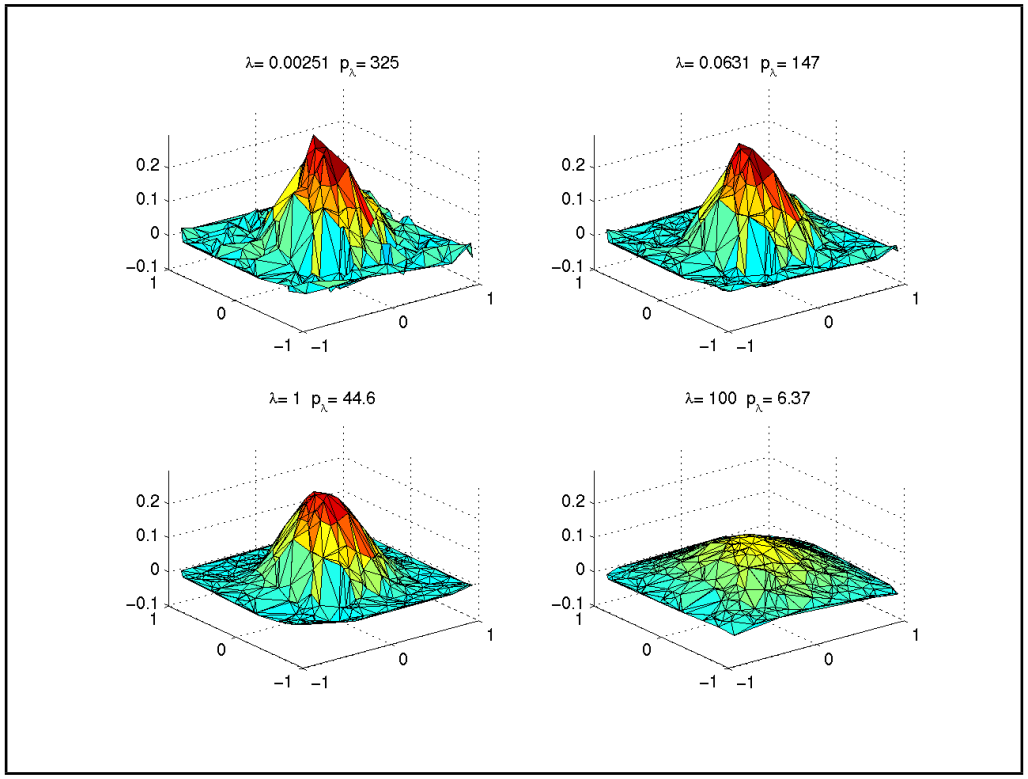


Figure 3: Four mean triogram fits for the noisy cone example. The values of the smoothing parameter λ and the trace of the linear operator defining the estimator, p_λ are indicated above each of the four plots.

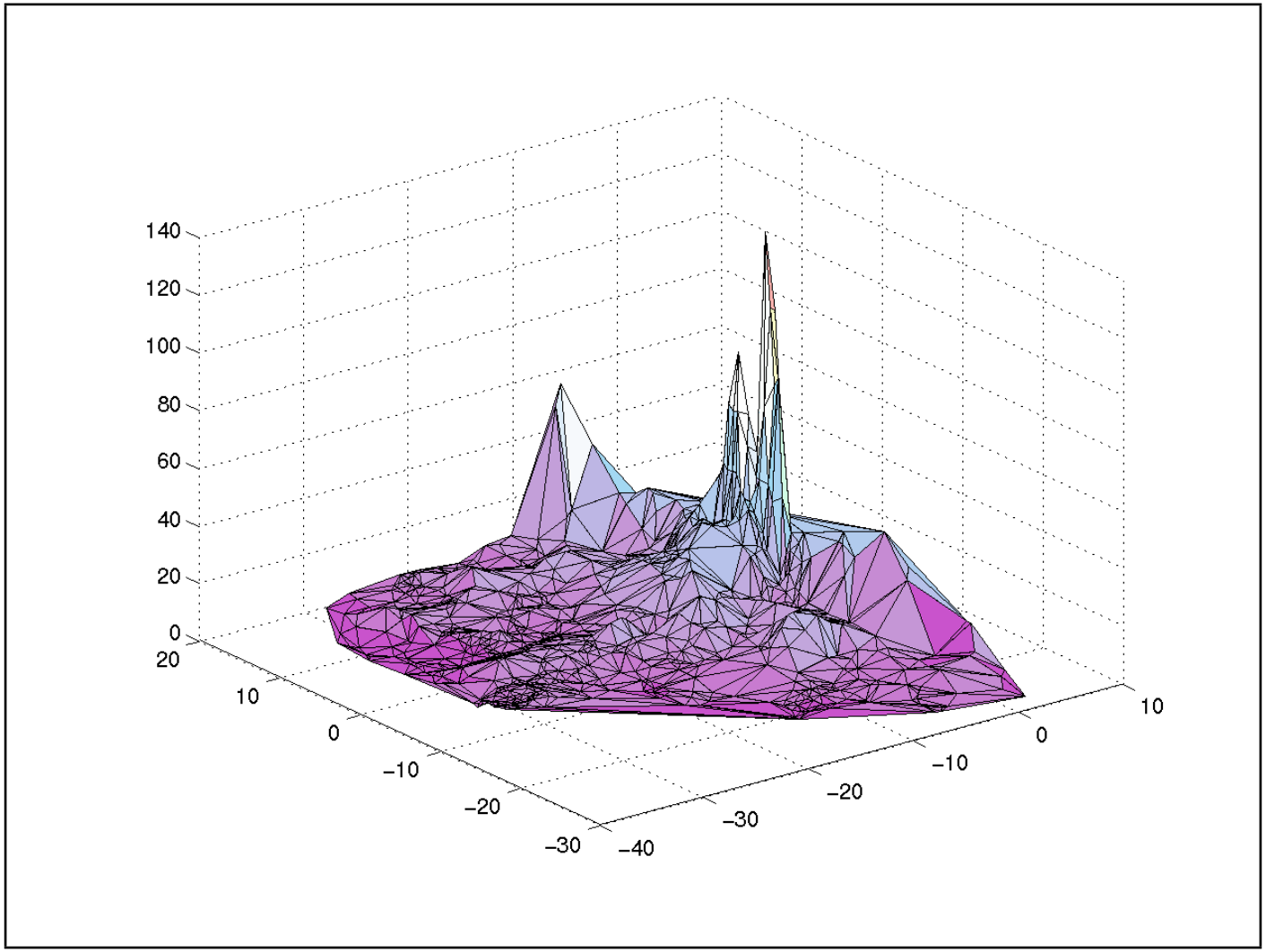


Figure 4: Perspective Plot of Median Model for Chicago Land Values. Based on 1194 vacant land sales in Chicago Metropolitan Area in 1995-97, prices in dollars per square foot.

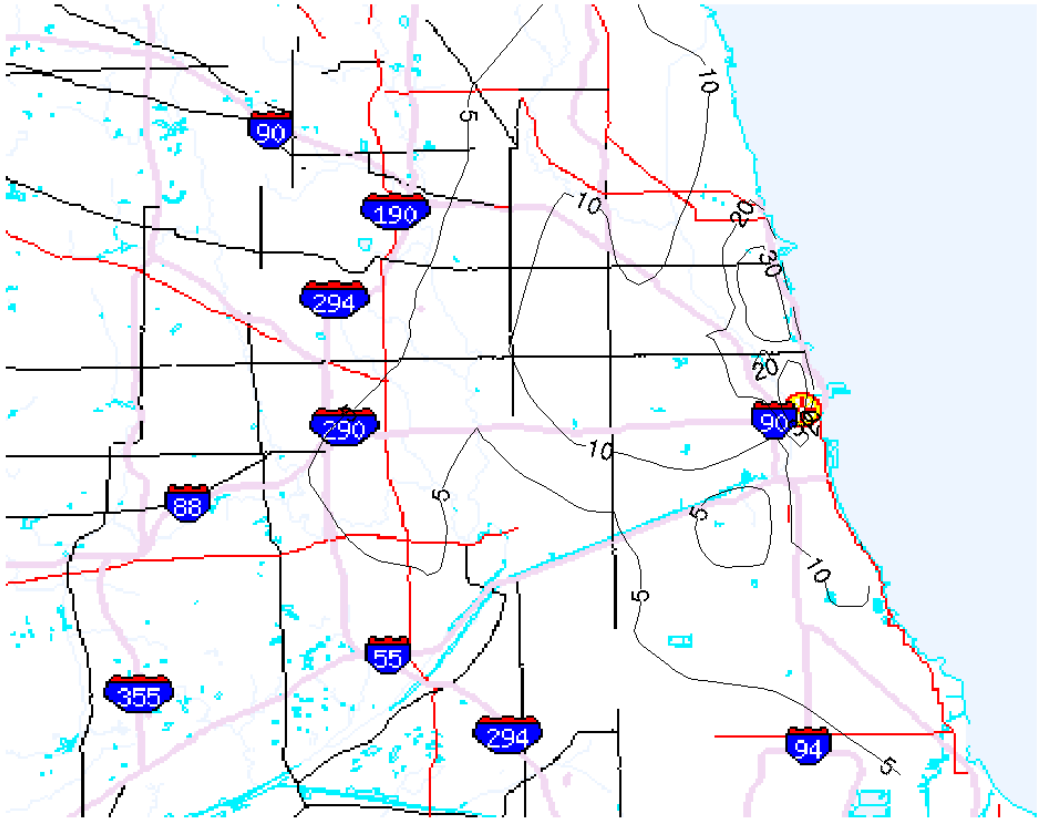


Figure 5: Contour Plot of First Quartile Model for Chicago Land Values.

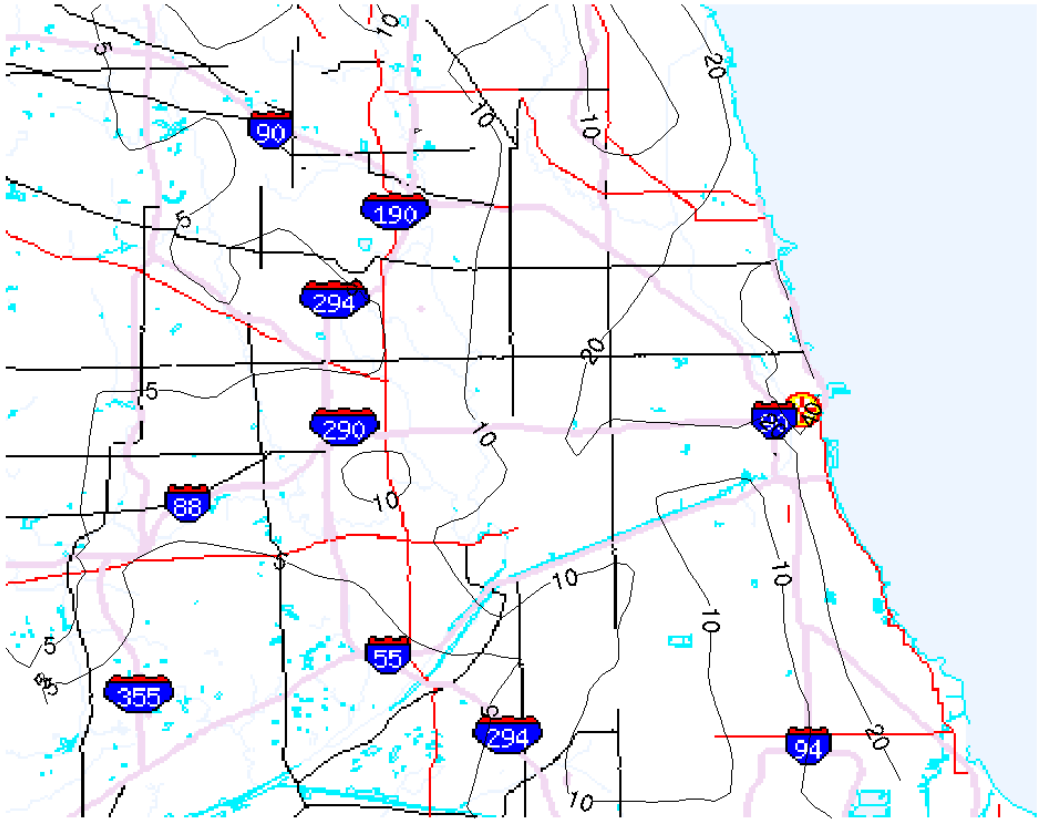


Figure 6: Contour Plot of Median Model for Chicago Land Values.

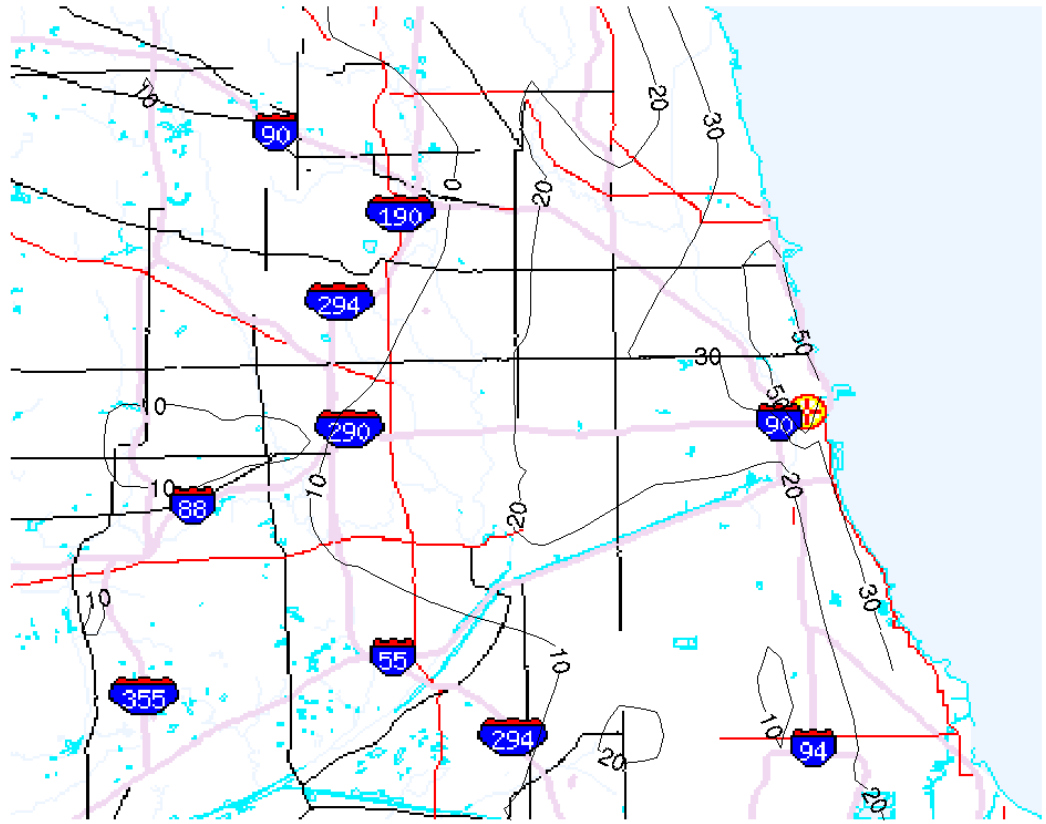


Figure 7: Contour Plot of Third Quartile Model for Chicago Land Values.

Automatic λ Selection

Schwarz Criterion:

$$\log(n^{-1}\sum \rho_\tau(z_i - \hat{g}_\lambda(x_i, y_i))) + (2n)^{-1} p_\lambda \log n.$$

where the dimension of the fitted function, p_λ, is defined as the number of points interpolated by the fitted function $\hat{g}_\lambda$. Other approaches: Stein's unbiased risk estimator, Donoho and Johnstone (1995), and e.g. Antoniadis and Fan (2001).

Extensions

Triograms can be constrained to be convex (or concave) by imposing m additional linear inequality constraints, one for each interior edge of the triangulation. This might be interesting for estimating bivariate densities since we could impose, or test (?) for log-concavity. Now computation is somewhat harder since the fidelity is more complicated.

Partial linear model applications are quite straightforward.

Extensions to penalties involving $V(g)$ may also prove interesting.

Monte-Carlo Performance

Design: He and Shi (1996)

$$z_i = g_0(x_i, y_i) + u_i \quad i = 1, ..., 100.$$

$$g_0(x, y) = \frac{40 \exp(8((x - .5)^2 + (y - .5)^2))}{(\exp(8((x - .2)^2 + (y - .7)^2)) + \exp(8((x - .7)^2 + (y - .2)^2)))}$$

with (x, y) iid uniform on $[0, 1]^2$ and u_i distributed as normal, normal scale mixture, or slash.

Comparison of both L_1 and L_2 triogram and tensor product splines.

Monte-Carlo MISE (1000 Replications)

Distribution	L_1 tensor	L_1 triogram	L_2 tensor	L_2 triogram
Normal	0.609	0.442	0.544	0.3102
	(0.095)	(0.161)	(0.072)	(0.093)
Normal Mixture	0.691	0.515	0.747	0.602
	(0.233)	(0.245)	(0.327)	(0.187)
Slash	0.689	4.79	31.1	171.1
	(6.52)	(125.22)	(18135)	(4723)

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Dogma of Goniolatry

- Triograms are nice elementary surfaces
- Roughness penalties are preferable to knot selection
- Total variation provides a natural roughness penalty
- Schwarz penalty for λ selection based on model dimension
- Sparsity of linear algebra facilitates computability
- Quantile fidelity yields a family of fitted surfaces